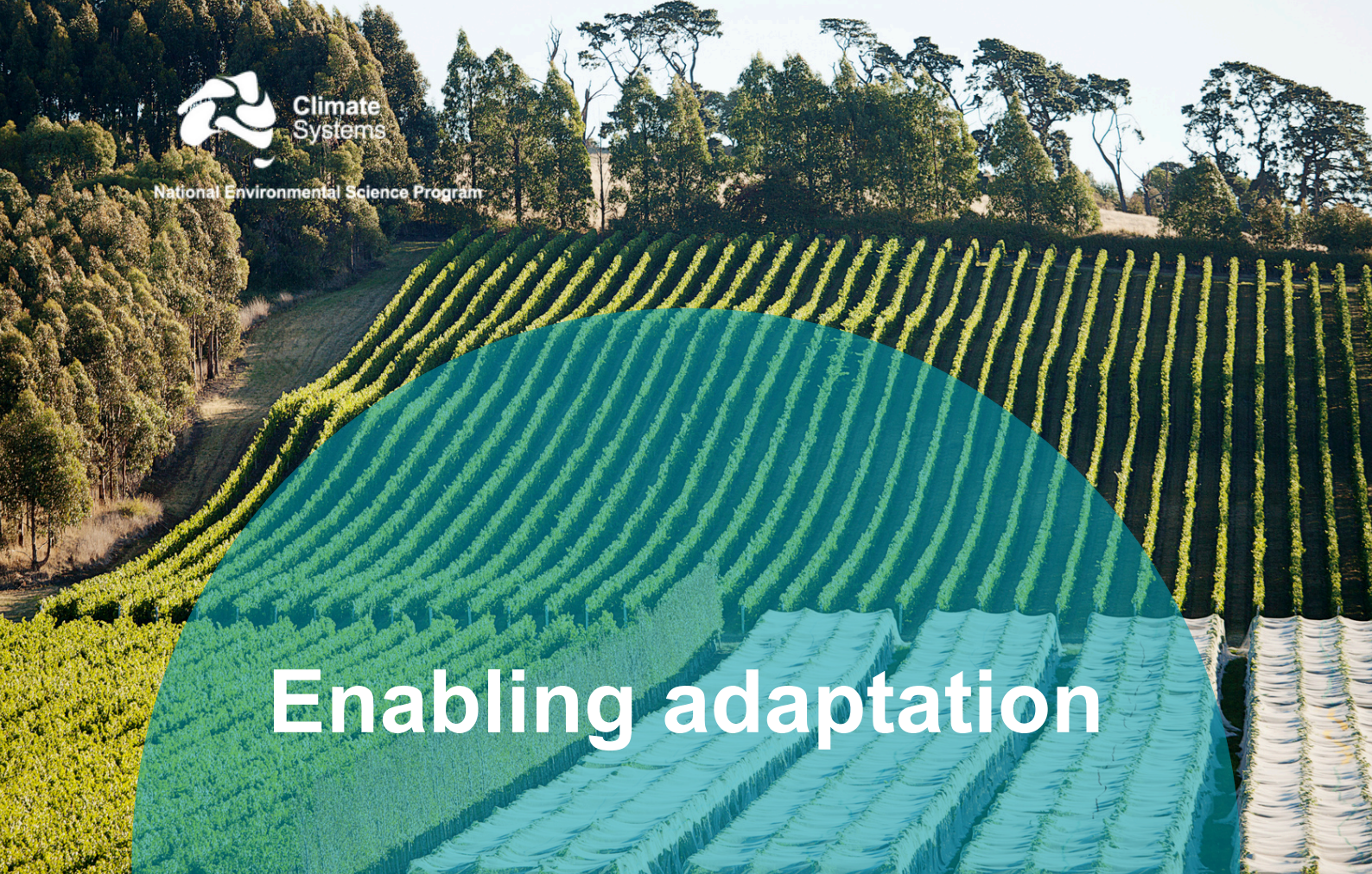




Climate
Systems

National Environmental Science Program



Enabling adaptation

**Enterprise Suitability
Mapping and
Tasmania's wine
sector**

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The Climate Systems Hub acknowledges the Traditional Custodians of the land across Australia where this work occurred. We pay our respects to Elders past and present and recognise the important role traditional knowledge plays in understanding Australia's climate.

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Cover image: Slovegrove Photos

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1. Introduction

For over ten years the Tasmanian Government of has been working to develop the state's agriculture industry. They have worked closely with industry partners to understand their needs, and invested heavily in infrastructure, knowledge and information products. Tasmania's statewide [Enterprise Suitability Maps](#) (ESM) is one product of this work.

The ESM are geospatial information products developed by the Tasmanian Department of Natural Resources and Environment (DNRE). They combine high-resolution digital soil mapping (DSM), climate information and crop suitability parameters to inform farmers, industry and investor decision making.

These ESM are considered a very successful decision support product. The National Environmental Science Program (NESP) Climate System Hub project Enabling Adaptation Best Practice set out to unpack what makes this climate information product 'enabling' of climate change adaptation. We focus on the rapidly expanding wine sector of Tasmania, one of the most frequent users of the ESM, to understand the influence of this climate information has had.

This case study traces a web of enabling factors, ranging from those that enabled the ESM product, through to those factors that enabled adaptation decision making and transformation by winemakers. In addition to discussing the value of information products which are developed in close collaboration with stakeholders and users, we review the necessity of other enabling factors alongside knowledge products.

2. Research methodology

This case study was undertaken between July and November 2024. Data was collected primarily through semi-structured interviews that focussed on process tracing to identify key enabling factors, and exploring why each project was considered good practice. Six interviews were conducted with implementors, consultants, wine growers and other stakeholders, as outlined below in Table 1.

Table 1: A summary of the participants interviewed.

Role	Number of participants
Implementor	2
Consultant	2
Wine grower	2

Online searches were used to collect additional information, existing academic research and any documents referred to in the interview process. As a result, 27 government reports and documents, GIS data, news reports and websites were included in the evidence for this case study, as well as 12 academic papers. Table 2 gives further detail on the supplementary data that was collected.

Table 2: A summary of supplementary data used to further inform the case study.

Type	Count
Government reports, policies and strategies	14
News report	3
Other reports	3
Website	3
Presentation	2
GIS layers (via DNRE)	2
ListMap usage data (via DNRE)	1
Academic papers	12

ENTERPRISE SUITABILITY MAPPING PROJECT TIMELINE

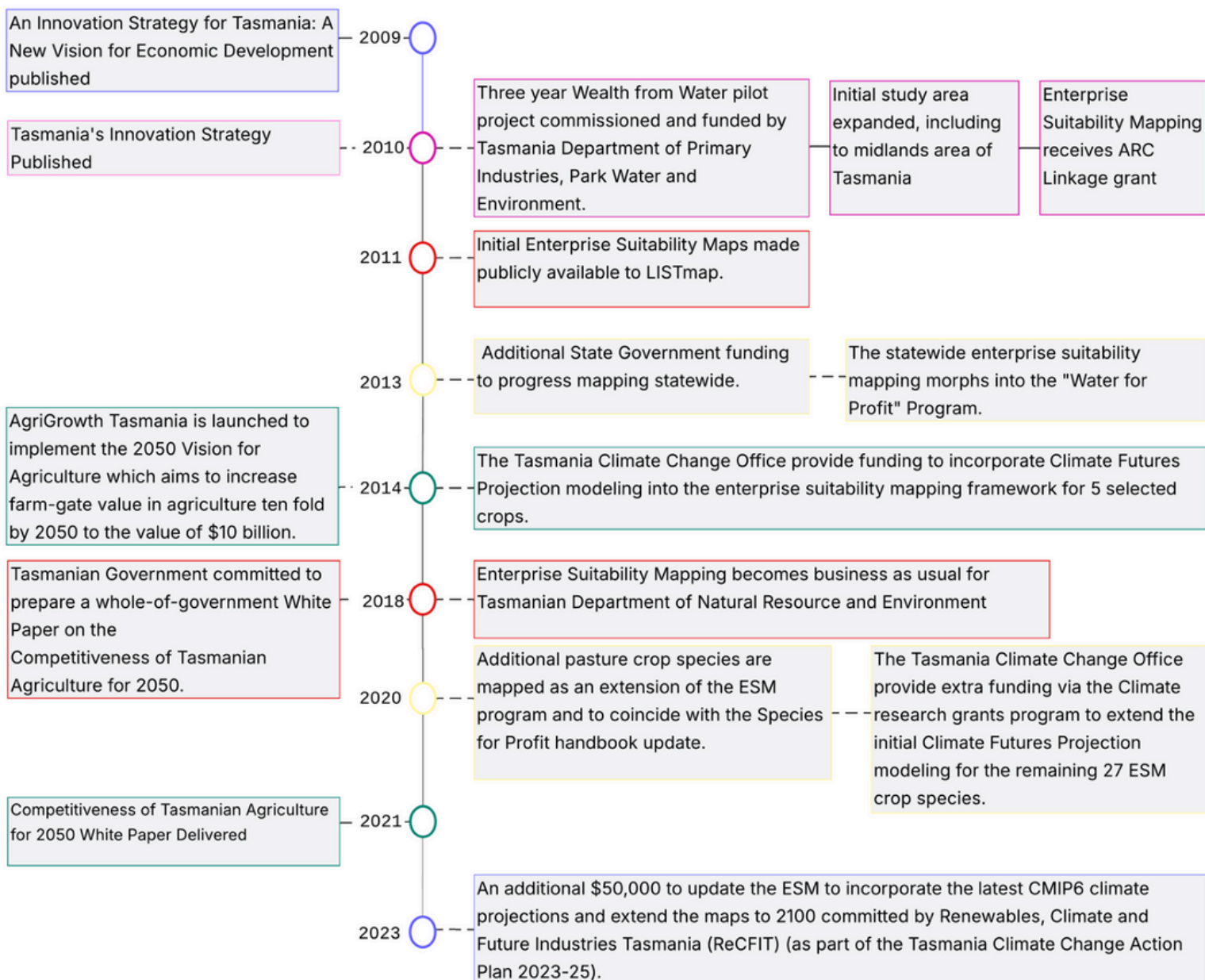


Figure 1: A timeline of major developments in the ESM project alongside major announcements from the Tasmanian government.

3. Background information

In 2009, in collaboration with the Tasmanian Government, the University of Tasmania published a report titled 'An Innovation Strategy for Tasmania: A New Vision for Economic Development'(1). This report explored the agricultural potential of Tasmania and was used to develop the Tasmanian Innovation Strategy (2). Together these documents highlighted Tasmania's agriculture industry as a high-value economic opportunity which needed investment in necessary resources and development. Irrigation expansion was seen as a priority for investment.

In 2014, AgriGrowth Tasmania was launched, which is a specialist group in the Agriculture and Water unit of the DNRE. This new group was designed to oversee Tasmania's Agri-Food Plan and related policies, and support the new commitment to sustainably grow the farm-gate agricultural value of Tasmania tenfold by 2050, to \$10 billion.

The Tasmanian Government has now developed 19 operational irrigation schemes as part of their plan to grow farmgate value by 2050. Investment into agriculture remains significant, with \$5.1 million committed to the Tasmania Institute of Agriculture in 2023-24 (3). In August 2024 they announced an additional \$1.6 million for research into irrigation efficiency and helping farmers adopt practices that minimise environmental impacts (4,5).

These examples of the significant policies and investments made by the Tasmanian Government in recent years demonstrate the political commitment to develop a combination of infrastructure, knowledge products and capacity development activities to support growth and change in Tasmania's agricultural production.

a. What is the Enterprise Suitability Mapping project?

The ESM is an integral part of the development of Tasmanian agriculture. It was initiated through a three-year Wealth from Water pilot project, commissioned and funded by Tasmania's Department of Primary Industries, Park, Water and Environment (DPIPWE), in 2010 to understand land suitability and stimulate investment alongside new irrigation schemes (6,7). After initial scoping, an Australian Research Council Linkage Project was developed between the University of Sydney Faculty of Agriculture and Environment, DPIPWE, and the Tasmanian Department of Economic Development, Tourism and the Arts , Institute of Agriculture and the Australian Collaborative Land Evaluation Program (6–8). The project commenced as a pilot study of 43,000ha in Tasmania's Meander Valley and later extended to include an additional 27,000ha in Tasmania's Midlands (7).

b. An overview of the Enterprise Suitability Maps

The maps are provided in an online [LISTmap portal](#) as a decision-support tool for farmers, consultants and investors (8). The maps provide guidance on:

1. Areas where crops could be introduced, intensified or diversified
2. Possible risks or impediments to growing crops
3. The viability of new enterprises at a state or regional scale.

The maps are available along with additional information, methodology and supporting metadata (9). They are built for a variety of stakeholders, catering for both a simple overview of conditions or more in-depth analysis with downloadable data.

The spatial data product allows users to query any part of the state-wide map, by simply entering a street address, to view suitability class, soil, climate and terrain parameters and information on vulnerabilities or risks (8). Market tools, financial modules, irrigation case studies and fact sheets also exist alongside the map products to inform management requirements for an enterprise to become profitable (6,8).

The maps provide information to preliminarily identify areas that could be used for new enterprises, intensification, or diversification into irrigated land, for further investigation. They could also be used to guide policy development, including land use planning and the allocation of agricultural land (6).

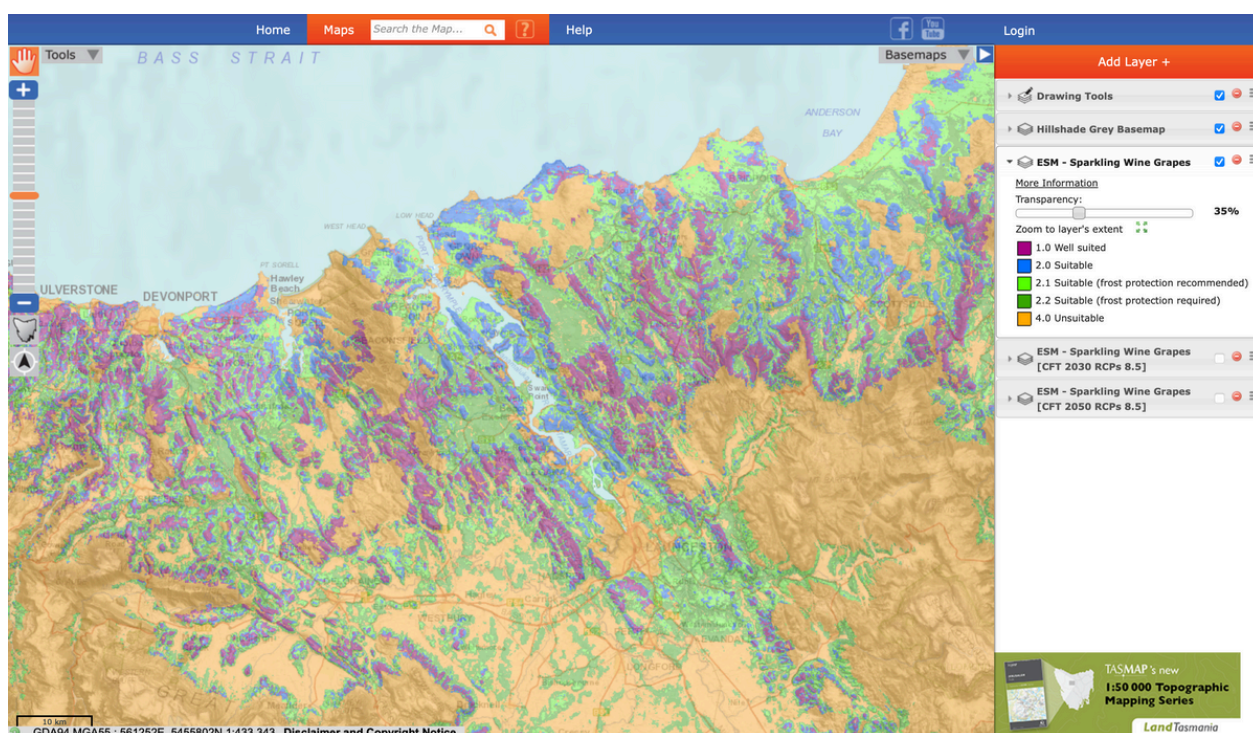


Figure 2: A snapshot from the LISTmap Portal.

c. Longevity of the Enterprise Suitability Maps

In the 3 months following their release there were over eight thousand viewings of the maps (6). The Tasmanian government therefore saw value in investing additional funds, and in 2015, the Water for Profit program was launched, which aimed to ensure farmers were equipped with the right skills and information to maximise their return on investment for irrigation by extending the project to state-wide mapping at 80 m resolution (6). This was later downscaled to 30 m resolution in 2018 (8).

In 2015, the Tasmania Climate Change Office provided additional funding to begin testing the incorporation of Climate Futures Projections modelling into the ESM. In 2020 this was extended with additional funding via the Tasmanian Climate Research Grants Program to cover all crops.

The Water for Profit program concluded in 2018, however the ESM continue to be maintained and updated as business as usual for DNRE. Current work on under Tasmania's Climate Action Plan 2023-25 includes extending the maps to 2070 and 2100 and provide new [real-time climate modelling](#) information products to inform how each season is tracking.

d. Method for developing the Enterprise Suitability Maps

The methodology for developing the ESM project is documented extensively in existing academic literature, as well as in online metadata for the products (8,7,6,10–15).

The maps are created using a model based on Digital Soil Mapping methodology. The digital mapping is a more cost-effective approach to scalable soil mapping, which had not been tested in an operational context of this scale prior to the pilot study (8).

'Suitability rule sets' developed by the Tasmanian Institute of Agriculture summarise the ideal soil and climate requirements, and major limitations, for common crops in Tasmania. This included vegetables, cereals, pastures and forestry (7,9). The term 'enterprise' was used to describe the product because these rules also take into account the level of management required and other business parameters, including gross margins analysis and opportunities analysis (6).

Soil samples were collected from across the project site and temperature sensors installed in optimum locations to generate climate data (6). This includes data on soil property, pH, clay content, stone content, and drainage class (7). Climate parameters include frost risk, heat risk, extreme rainfall risk, chill hours, or the number of 'growing degree days' (i.e. suitable temperature days for growth and ripening).

Together, the soil and climate data were fed into the model and analysed against the suitability rules to produce suitability scores for each crop in a given location. These maps are produced at 30m resolution.

After the initial pilot, the methodology was extended to incorporate climate projections data. Using an initial sample of five crops (barley, poppies, potatoes and wine grapes) (sparkling and table wine), current climate data in the model was replaced by climate projection data from the Climate Futures for Tasmania dataset. These layers demonstrate the suitability of crops under future climate scenarios. This was later extended to cover the remaining 27 ESM crops.

Each of the 32 crops now has three map layers available pertaining to current climate, near future climate (2030) and future climate (2050)(9). A versatility index has also been created that summarises the number of crops that are well suited to an area.

A paper by Kidd et. al. gives an idea of the total resources that were required for the initial 80,000 ha pilot study and subsequent state-wide sampling and modelling, including 8 full time staff and 270 temperature sensors (8).

e. Industry consultation and user engagement for the Enterprise Suitability Maps

Industry collaboration and stakeholder engagement was a priority for this project.

The suitability rule sets were developed based on industry workshops, existing literature and input from agronomy experts. Crops were prioritised and selected for inclusion through workshops, technical working groups and reference groups with key stakeholders.

Multiple formal rounds of consultation and feedback gathering have been conducted to improve the product usability, as well as more ad-hoc conversations with key stakeholders to tweak the products. Face to face workshops with different industry stakeholders were conducted, and included demonstrations of the map products. LISTMap training sessions continue to be held every year as part of the Tasmania Agriculture Productivity Group Expo. Additional trainings are held with Natural Research Management Groups, consultants, and other users upon request.

f. Climate challenges facing Australia's wine industry

Climate change is expected to affect all aspects of the wine industry, and there is increasing evidence the sector is already being affected (16). Some of the conditions which cause major concern include:

- extreme heat
- changes in temperature patterns
- changes in rainfall patterns, humidity and moisture
- drought and uncertain water supplies
- flooding and waterlogging
- hail
- frost
- bushfire smoke taint
- fruit sunburn or shrivelling.

These climatic conditions can have a significant impact on disease, pests, vine health and ripening times. Ripeness also impacts the sugar content, colour, flavour and alcohol content of a wine (16,17). On top of this, changes in harvest times are increasing logistical challenges and putting pressure on the infrastructure required for harvest, as well as winery processing equipment and tank space. Fruit pickers and other workers face additional occupational health and safety concerns from working in extreme heat conditions (16,17).

Australian wine growers are accustomed to the various climate challenges that can arise from different seasonal patterns. However, with changes in temperature and rainfall, and increases in the frequency and severity of extreme events, these climate challenges become increasingly destructive.

g. Adaptation options for wine growers in Australia

Historic experience managing climate variability means the wine sector, and agriculture more broadly, has been well placed to build on existing knowledge and skills to support climate change adaptation. The wine sector is already implementing and testing an impressive number of different adaptation techniques.

Location

Winemakers are beginning to source grapes from, or grow grapes in, cooler regions or high altitudes to reduce heat stress (although high altitude areas often face increased risk of smoke taint from bushfires). It is for this reason that Tasmania, which has Australia's coolest climate, has been of increasing interest. With increasing irrigation infrastructure, the amount of suitable land in Tasmania for wine growing is increasing. Tasmania also faces lower bushfire risk.

Varietals

Investing in new varieties of grapevine has also gained significant attention in recent years. This can include later or earlier ripening grapevine varieties, importing new varieties from hotter climates, or developing and trialling new more heat-resistant varieties (as has been done by CSIRO (18)). Some vineyards with existing infrastructure are 'retrofitting' or replanting with these different varieties.

Technology

Real-time monitoring of climate conditions is used to identify and mitigate risk factors. Adoption of technology into the vineyard management can include weather, plant and soil sensors, satellite and remote sensing (16,17). New technology is also used in site selection, vineyard layout and design.

Vineyard Structure

Different trellis systems are used to manage canopy and shade. Tasmanian vineyards like Apogee are using the 'Scott-Henry' trellis system to increase leaf area to capture sunshine(19). Canopy can also be used to increase shade and protect the grapes during heatwaves in warmer climates.

Changing row orientation from the traditional north-south rows to east-west rows to reduce sun exposure is becoming increasingly common in areas like Victoria. Some older vineyards have been completely re-designed.

Other adaptation techniques include changes to irrigation practices to increase efficiency, protecting fruit from extreme heat and sunburn, changes to harvest time or changes to winemaking practices.

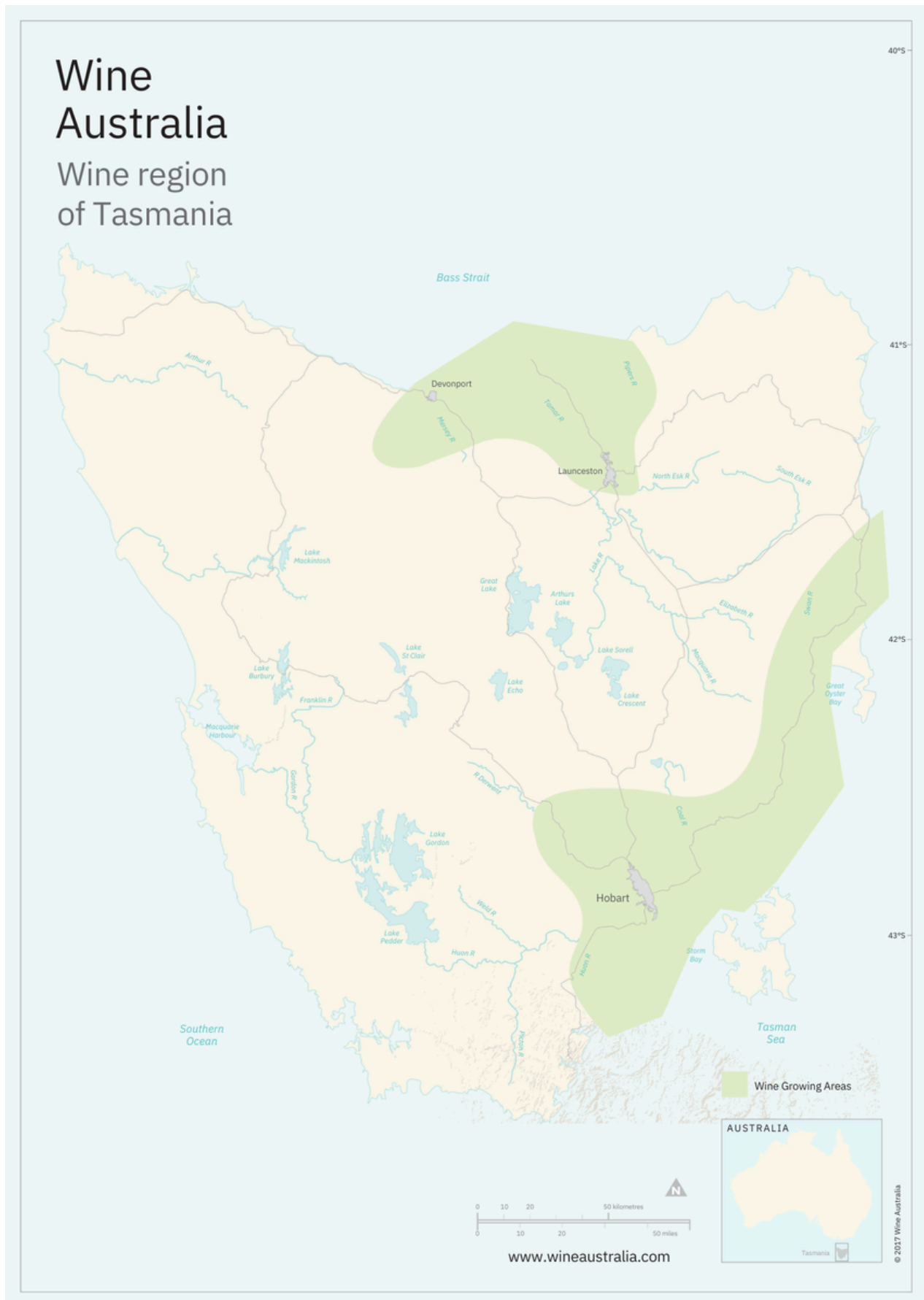
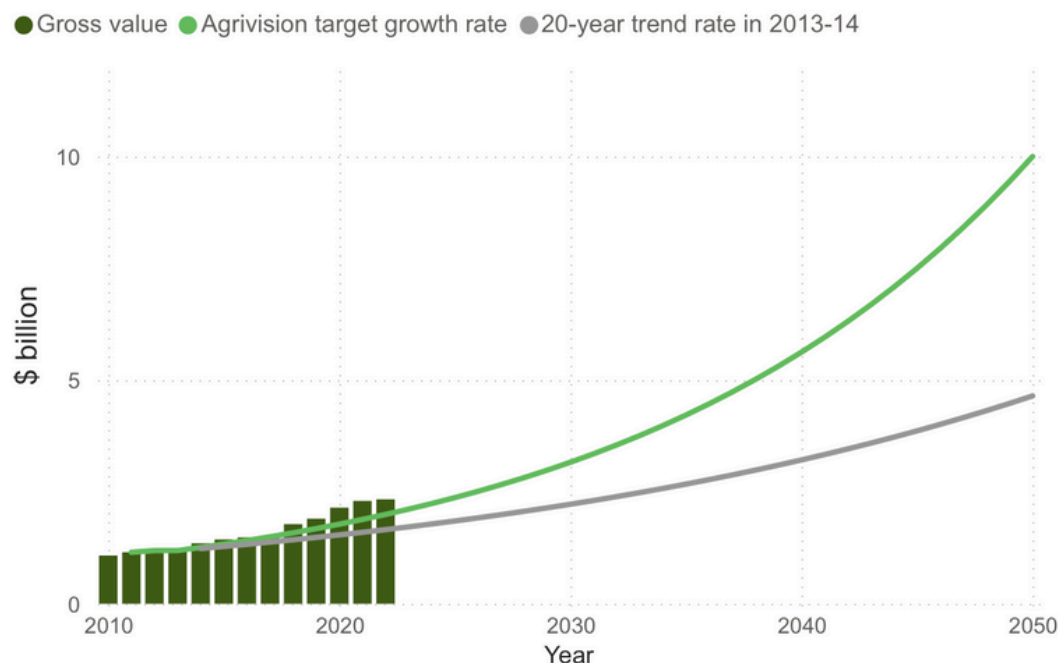


Figure 6: A map showing Tasmania's wine growing regions (Wine Australia)

4. Findings

a. Growth in Tasmania's wine industry



Further information on the Agri-food sector and progress against the AgriVision 2050 target is outlined in the Agri-Food ScoreCard. The ScoreCard is an annual Agri-Food sector statistical report. The latest ScoreCard report can be accessed by clicking the link

Figure 3: A graph from the Tasmania Agri-Food Scorecard showing the increase in Tasmania's agriculture farm gate value since 2010 (Tasmania Agri-Food Scorecard Dashboard).

Current statistics demonstrate Tasmania is on track to achieve the desired increase in agricultural production. Farm gate value has increased from 1.08 billion in 2009-10 to 2.34 billion in 2021-22 (Figure 3) (20).

One significant contributor to increasing farm gate value is Tasmania's wine industry, which has increased substantially from \$17.92 million in 2010-11 to \$43.12 million in 2021-22, as shown in Figure 4 (20). The map below (Figure 5) also visualises the increasing number of vineyards in Tasmania's southern wine regions from 2009 to 2021.

The data contained in these graphs comes from the Tasmania Agri-Food Scorecard Dashboard, which uses Australian Bureau of Statistics data¹.

[1] Australian Bureau of Statistics, Agricultural Commodities, cat. No. 7121.0, ABS Value of Agricultural Commodities Produced, cat. No. 7503, Trade Data (international food exports and imports) and industry data from entities such as Wine Tasmania, Wine Australia, Horticultural Innovation Australia etc.

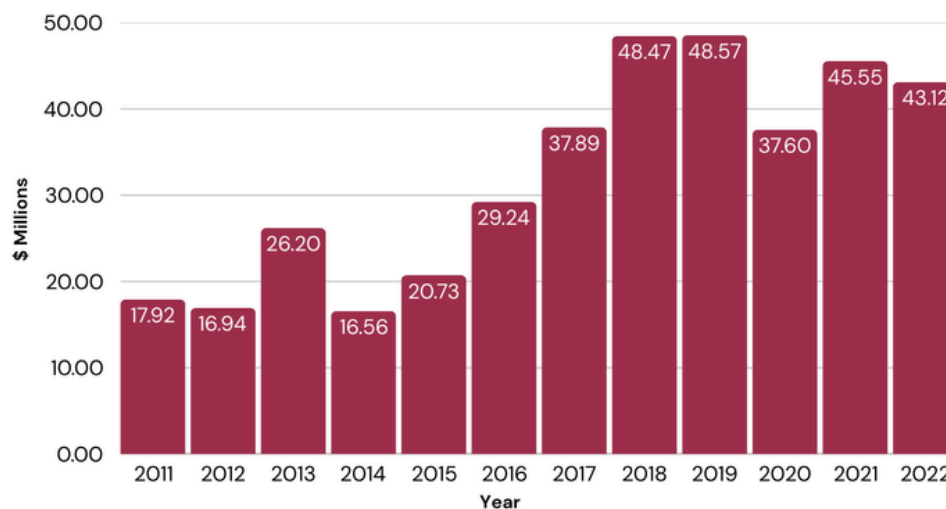


Figure 4: A graph adapted from the Tasmania Agri-Food Scorecard showing the increase in Tasmania's wine production farm gate value since 2010.

This growth in Tasmania's wine industry is timely and well aligned with both a new-found recognition for Tasmania's value as a cooler climate growing region, and the increasing investment in new irrigation infrastructure and industry support.

The increase in production has also come with concerns of wine oversupply and loss of value, given the Tasmanian wines were previously a premium product selling for higher prices (3).

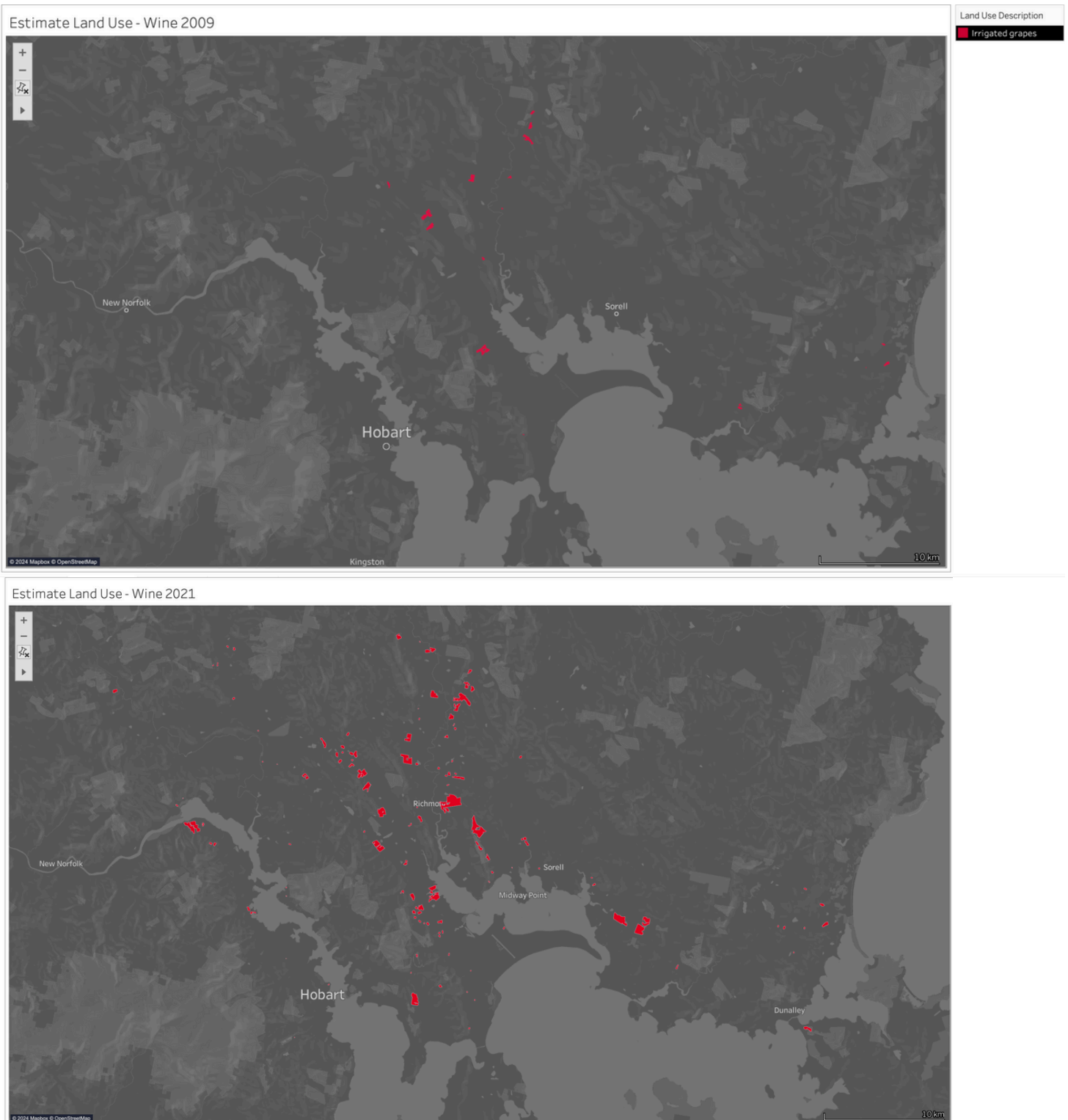


Figure 5: Land use mapping showing the increase in land used for irrigated grapes between the years 2009 and 2021 for Tasmania's Southern wine-growing areas (DNRE).

Example 1: Brown Family Wine Group Moving South

Brown Family Wine Group is a family-owned business based in northeastern Victoria, Australia. Their primary label, Brown Brothers, is one of Victoria's oldest and well-known wineries, dating back to 1885. They are now in their fifth generation of wine growers.

Brown Family Wine Group began to observe an increasing impact of climate on their production in the early 2000s during the height of the millennial drought. Limited water supplying the Murray Darling Basin placed significant strain on the business and forced them to take on huge additional costs purchasing water, as well as significant losses. Smoke taint from nearby bushfires also caused significant crop damage. Although retail shelf space was reduced and sales faced challenges during this period, the experience created opportunities to innovate and adapt for future success.

Their experiences, coupled with increased discussion about climate change at the time, had brought climate to the attention of senior leadership. In 2008, the Victorian Department of Primary Industries (now Department of Environment and Primary Industries) was invited to present to the board and senior management team about the future impact of climate change. Together, this led to a board resolution to assume a 2-degree temperature increase by 2050, ensuring that climate change was factored into the decision-making process for all capital projects.

“It became very clear that what we had experienced was a taste of what was to come.”

Brown Family Wine Group began risk assessments on their existing vineyards, analysing how they were expected to cope in the future. Various adaptation responses started to be built into business as usual, such as more efficient irrigation practices, and the planting of heat-tolerant species. Safeguarding cooler-climate grape varieties against rising temperatures proved to be a greater challenge.

It was a timely opportunity when Brown Family Wine Group were contacted about purchasing three vineyards in Tasmania's Tamar Valley. Tasmania offered opportunity for expansion into a cooler climate region and new markets. With the backing of the company, this purchase was finalised in 2010.

“All the signs kept pointing toward Tasmania at that point in time.”

Investing in Tasmania was quickly seen as a success. The cooler climate varieties, such as Pinot Noir, sold well and offered a worthwhile addition to the brand. Brown Family Wine Group made significant investments in Tasmania's infrastructure, focusing on planting new vines and forming partnerships with landowners interested in becoming growers. Brown Family Wine Group has become well known for these adaptation responses to the climate challenges they have faced. The decision to invest in vineyards in Tasmania is one which grabbed the media's attention and was discussed quite widely, becoming a part of their brand reputation.

b. Evidence of wine growers using the Enterprise Suitability Maps

Widespread use of the ESM product by Tasmania's wine growers suggests the government's investment has had a positive influence on growth in the wine sector. Data from the DNRE show that the 'Table Wine Grapes' 2018 layer remains the most popular ESM, averaging more than 20,000 hits (number of clicks) each month. Across the current and future climate layers, the 'Table Wine Grapes' ESM made up an average 10% of all usage in 2024.

Evidence from interviewees demonstrates the value winegrowers have found in the ESM products.

Agricultural consultants are regularly contracted to identify and develop suitable land, inform new investments and conduct due-diligence checks. These consultants have used the maps as an initial guide to inform further analysis:

“I think we are very lucky to have all that data at our fingertips, and it's really easy to access. It's easy to use. You've got the data, and you can pull it into your own systems and interrogate it.”

Many consultants also discussed 'ground truthing' with additional data (such as Lidar) to create similar products at a finer scale. The ESM can also be aligned with Bureau of Meteorology (BoM) data and other existing datasets.

The information contained in the ESM can be used to recommend choice of vine species, design of a vineyard, or other risk management (adaptation) strategies that might need to be planned for. Consultants highlight that the frost risk maps, and growing degree day maps are the most important products because they address the two most significant climate risks for growers in Tasmania and this information is not available elsewhere:

“...the frost risk maps are fantastic... that's information that I can't process or get ... They've also got growing degree days. That's very handy.”

Thanks to the extensive data that sits behind the maps, the ESM are believed by interviewees to be accurate in predicting these climate risks.

Evidence from this case study also suggests growers use the ESM products themselves to inform decision making. Here it is important to understand landowners and growers can range from small hold landowners to large scale enterprises with multiple properties of interest. Both types of growers have made use of the maps to gain a high-level understanding of their land, although their ability to drill down further depends on experience with GIS and technical capacity. Brown Family Wine Group have used the ESM in their climate adaptation journey outlined above (example 1) including to compare locations, help choose suitable grape varieties and identify new land for investment. The example from Dalness Estate (example 2) also demonstrates a case where the ESM were used to direct new development.

The user design and accessibility of the product is extremely important to their success. The LISTMap portal is a simple and easy to use way of interacting with data that can be accessed by a range of skillsets. Being able to enter a street address to begin searching data on standard map structures is something users regularly discussed as useful.

Anecdotal evidence from interviewees suggested some cases where the maps have also been used for conservation purposes and by Natural Resource Management (NRM) groups, including using the versatility index maps to inform suitable land use planning and identify suitable conservation areas.

Example 2: Using the ESM for new vineyard development at Dalness Estate

Dalness Estate is a large historic property over 1700 hectares. In 2017, the estate contracted a consultant to identify a suitable location for a new vineyard on the property. The ESM maps were used to narrow down areas on the property that could be suitable and analyse potential risk factors.

Two suitable locations were identified: one was cheaper to develop but slightly higher risk for wine growing, and the other was more expensive to develop due to its location and terrain. The owner chose to develop the more affordable site, but after initial success proceeded to develop the second site.

Several years later, both vineyards are performing well and have not suffered from frost (though it is likely the higher-risk site will on occasion). In total, they now have developed close to 40 hectares of vineyard and investment is estimated to be greater than \$4 million.

This is a clear example of a project which has used maps in decision making and future planning for a vineyard, which also led to a positive impact on the surrounding district, including an increase in local employment.

“The suitability layers were very critical because without that support you wouldn’t know where to go on such a big property.”

c. Recognising barriers as context to enabling factors

The project team analysed interviews and documents to better understand some of the factors enabling both the ESM project and the adaptation taken by wine growers. Understanding some of the significant barriers that were overcome provides important context to these enabling factors. While enabling factors are not necessarily the inverse of barriers, evidence suggests they do assist in circumnavigating barriers.

The most significant barrier faced by the developers of the new methodology for DSM and ESM was scepticism about the accuracy and usefulness of the product. The agricultural industry was loyal to traditional soil sampling and mapping, a very resource intensive, expensive and subjective exercise that is difficult to execute state-wide. This is discussed by Kidd et al. (8). Alongside the scepticism, additional work was required to get industry onboard with committing the significant investment of time and resources that was needed for quality engagement.

As is always the case, this project faced the challenge of limited resources. There was limited computing power available within Tasmanian Government for producing such a product. While the ESM methodology reduced on-ground data collection, the project still faced some of the challenges of resource intensive data collection and soil sampling. Developing this product also required huge knowledge inputs and capacity building, for example learning to code and develop machine learning was a '*big learning curve*' for staff involved.

Wine growers looking to invest in developing vineyards in Tasmania also face a number of barriers to entry that need to be overcome. Developing entirely new infrastructure and planting requires high start-up costs. This is less so for mainland Australia where vineyards can be purchased. Tasmania's cool climate and climate variability also brings its own risks which must be managed. Irrigation infrastructure has only been developed in recent years and water shortages still exist on occasion. Finally, with the growth in Tasmania's wine industry there's a concern market saturation will reduce the value of their product and lessen return on investment.

d. Factors enabling the Enterprise Suitability Mapping

Enabling Factor:	Explanation
Institutional support	Tasmanian Government support as part of their wider agricultural investment strategy played a huge role in enabling the ESM project. The pilot proof-of-concept was essential for the support of both the state government and industry. It helped get people onboard and gain investment. Initial concerns about the product had to be overcome by producing quality evidence through academic peer-review, as well as by sharing the methodology to make the product transparent to users (8).
Industry support	Industry support was of equal importance to the ESM. Industry partners and growers were the main beneficiary and stakeholder, and their support and use of the final product was necessary to justify investment.

Enabling Factor:	Explanation
Sufficient resources	The ARC linkage grant was an essential enabler, providing the necessary resources and opportunity for collaboration. Nectar Research Cloud and other computing resources which became available through collaboration with the university was essential to filling resource gaps. Working with the university also brought additional funding and capacity development opportunities that are often not available through government channels. Securing ongoing funding as part of 'business as usual' has been key to the sustainability and success of the ESM. This has been possible through alignment with the Tasmanian Government's strategic direction.
Efficient use of resources	Alongside the importance of resources comes the importance of using those resources efficiently. For example, during the project implementation a statistical approach to placing the sensors identified that air temperature sensors from the initial pilot could be dispersed further apart to cover the entire state without a need for additional equipment. This new methodology was a much more cost-effective approach and allowed for "the best bang for buck in terms of monitoring climate long term across the site." Sensors were also aligned with BoM stations to draw upon long-term averages. The team also reflected on having to "scrimp" off resources they were able to access through their association with the university.
Collaborative capital	Collaboration was viewed as essential to the success of the ESM. Working with The University of Sydney allowed access to resources and knowledge and extended the project scope. It also allowed for collaboration with academics around the world. Collaboration with the CSIRO and other jurisdictions (other states and territories) was also valuable for sharing information, resources and learning. CSIRO's support for the project validated the methodology and created greater momentum: "CSIRO has played a big role because they've helped fund a lot of the collaborations and national community catch-ups." A community of practice for DSM practitioners across the country also provided value, collaborating on ideas and methods (e.g. sharing code). Similarly, using open-source software and code was an important enabler because of the collaborative nature of the community and sharing of code, helping carry each other forward (8).
Participation	The engagement and consultation conducted for the ESM was crucial in gaining support from stakeholders and designing end-products that users valued (8). Importantly, interviewees highlighted that engagement and consultation was a big part of this project from the outset rather than being added on in later stages. Interviewees also recognised the value participation added to the project: "[A user] came to us and came up with some really good ideas, on how we can improve the products, and how we should be looking, moving forward."

Enabling Factor:	Explanation
Knowledge – academic	As a knowledge product, the role of building capacity and sharing knowledge was essential to the ESM project. The collaboration between research and practice allowed the Tasmanian Government to access new knowledge (8). The implementers were working across the Tasmanian Government and university to ‘bridge the gap’ between research and practice. Participants also reflected on the importance of going to conferences and engaging in research to understand new practices that were being developed: <i>“We were given the option to go to a fair few national and international conferences... Keeping up with the research world was really important.”</i>
Leaders	The personal commitment of implementers to spearheading and fulfilling this project was observed to have a significant impact. These individuals have been working on the ESM products over ten years and their commitment is demonstrated also by their undertaking of PhDs in order to upskill during the initial project development. Members of the University of Sydney research team and CSIRO were also champions driving forward this work and making use of their existing networks.
Innovation	The inclusion of groundbreaking research allowed for development of a more innovative solution. The new DSM methodology was developed to help address resource constraints and ensure needs were met, overcoming constraints of a conventional analytical approaches (8).
New technology	The online LISTmap portal was an important resource enabling end users to engage with the product. A move away from traditional paper map approaches allowed for provision of much more data to users at a much broader scale (8). The new methodology for developing this product also enabled it to be conducted at a much broader scale than previously.
Flexible approach	In translating the new DSM approach from academic to real-world application, a flexible approach was necessary to allow practitioners to respond to their environmental constraints (i.e. ability to change plans if they can’t access one paddock to install a sensor) (8). On an organisation level, the Tasmanian government allowed for flexibility in their ways of working when developing this project, affording employees the flexibility to work across the ARC Linkage Project and PhD during their employment.
Effective Communication	Making the final product user friendly was essential for buy-in and longevity, which is closely linked to engagement. The ListMap portal caters to a variety of skills levels. There’s also a wealth of supporting resources, information and online materials available to support users. <i>“The consensus from a lot of those groups was to try and keep it simple otherwise grower groups and farmers or consultants won’t use it.”</i>
Opportunistic	It was described as <i>“serendipitous”</i> that the implementers were in conversations with the University of Sydney and could ‘opportunistically’ make use of the Linkage Grant program to expand the capacity of the project.

The enablers and drivers of developing the DSM product are outlined in more depth by Kidd et. al. (8).

e. Factors enabling wine industry adaptation

Knowledge - Information products	The ESM are an information product which is highly valued by the wine industry in Tasmania. Several things stand out for this information product (many of which are enablers listed above) that make it valuable: • It was designed in collaboration with stakeholders and has a good understanding of their needs. • It is usable and easy to engage with, for multiple skills levels. • It is downscaled to a resolution that is relevant for landholders. • Users have freedom to download data and use it for their own needs and purposes. • It is seen to be accurate, valid and reliable data. • It aligns and complements other data sources (such as BoM data). The Wine Australia Climate Atlas is another information product which has been useful for informing the decision making of the wine industry. By far the most important reason that the ESM has been useful for wine growers, however, is that the information being provided was accurately and precisely targeted toward fulfilling a need for information. It allows them to inform future planning with climate in mind: <i>“...that’s quite handy to have to be able to look at what’s potentially going to happen down the track.” “As a wine industry [I] think we’re adequately resourced with, with a reasonable amount of information for individuals to make decisions about their own enterprises.”</i> Real-time weather monitoring was also discussed as essential. Weather apps and local weather stations provide weather tracking and notifications if there is increasing risk of certain pests and diseases, allowing growers to act quickly. The Tasmanian government are now working on providing these real-time information products as well. This is possible because of the skills and knowledge developed from ESM: <i>“Crop modelling goes hand in hand with [weather monitoring].”</i>
Planning	Participants also discussed the value and importance of good planning and management. An orderly approach to viticulture enables future planning and better practice.
Resources	The ESM was not the only enabling factor initiated by the Tasmanian Government. The new infrastructure and other investments made, significantly enabled the wine industries expansion in Tasmania. In particular, investment in new infrastructure and resources has helped reduce the barriers to setting up new enterprises and incentivised greater private investment with greater financial return.

Organisational support	The case from example 1, Brown Family Wine Group, demonstrates clearly that institutionalised or mandated support from within an organisation, which enables adaptation actions to progress, is just as influential in the private sector as has previously been discussed in research on state and local governments. The decision to move to Tasmania was opportunistic, however the board resolution had laid the groundwork and created an enabling environment that allowed the business to make use of opportunity when it arose, to push forward with a move into Tasmania. The strategic decision, and leadership demonstrated by their board, allowed Brown Family Wine Group to become an industry leader in climate adaptation, building on their reputation and brand name. Their decision was backed by a willingness to invest in strategies that enable them to better deal with climate change or build resilience. <i>“...an adoption of a policy at a board level followed up with actual action and then seeing it turn into a really commercial, successful commercial investment.”</i>
Risk perception	For winegrowers, one incentive for pursuing adaptation was observed to be an increased perception of climate risk. For Brown Family Wine Group, this was driven by experiences during the Millennium Drought, which incentivised decision-making and investment that takes into account future climate.
Values	The future-planning mindset of Brown Family Wine Group, and their concerns for the wellbeing of future generations, played a major role in their decision making. As a family business, Brown Family Wine Group were also able to take risks on investing based on values and good outcomes rather than solely short-term financial gain. Return on investment for adaptation can sometimes be slow and counter-intuitive, for example, if new irrigation infrastructure is not used until the next drought period. Even then, this might only result in avoided losses rather than increased revenue. <i>“The expectation is that future generations are going to have a viable business to work in over the next 130 years.”</i>
Opportunistic	For winegrowers, increased investment in Tasmania’s wine industry occurred after increased economic opportunity was created by new investment in infrastructure and resources. Brown Family Wine Group saw the decision to begin investing in Tasmania as ‘opportunistic’: the board decision and strategic direction laid the groundwork and set them up to act on the offer to purchase three existing vineyards in Tasmania.

5. Discussion

This case study sought to understand how an information product like the ESM can enable climate change adaptation to occur. We explored if and how information results in practical on-ground actions that reduce vulnerability to climate risks. It quickly became apparent that the ESM product was part of a much larger piece of work from the Tasmanian Government.

At the outset, evidence suggests the Tasmanian Government's strategic commitment to support the development of the agricultural sector of Tasmania has had the desired impact of promoting economic growth, at least in the wine industry. This can likely be attributed to the substantive actions taken to implement this policy: including ongoing political support and continuity in investment. Evidence also suggests that the policy has been well informed, with a good understanding of the gap in the market and significant funding for the development of knowledge or new information to support the investments being made.

This policy was also extremely well timed in that it aligned with increasing attention from interstate wine makers like Brown Family Wine Group who suffered due to the Millennium Drought. This opportunity was recognised in early reports from the Tasmanian Government, though it is difficult to attribute timing which could also have been a matter of good fortune. Indeed the scope of the ESM project was much broader than simply producing climate information. The project's outcomes can largely be credited to its accuracy in identifying and fulfilling a need of the agriculture industry. For the developing wine sector in Tasmania, accuracy in understanding industry needs resulted from careful engagement activities and a strong relationship between the Tasmanian Government and industry. This reflects the well-understood importance of proper consultation and engagement and provides an example of its value.

Recent research demonstrates an increasing understanding that not all enabling factors are equal. Differentiating between enabling 'conditions' which are ideal or necessary for adaptation but are not solely sufficient and 'catalysing' enablers which initiate or accelerate adaptation action is important (21). Tracing the connection between the ESM and on-ground adaptation action, it is unlikely the information itself is directly responsible. Other conditions and catalysts played a very significant role. The enabling environment was still conditional on several factors out of the control of policymakers, such as timing or incidence of climate change and perceived risk. Climate information can provide important guidance for adaptation decision making, but evidence from this case study suggests it as an enabling condition which works best when well supported by other enabling factors.

In Figure 7 we begin to visualise some of the complex enabling relationships between Tasmanian Government policy, the ESM project and the wine industry's adaptation decision making discussed here.

Of interest, we begin to notice how similar enabling factors can manifest differently in different contexts or with different actors involved. The idea of 'institutional support' for example, translated well in the case of the Brown Family Wine Group, where support from the board and organisation was significant. Similarly, while 'social capital' has been well recognised as important in local scale adaptation research, this project demonstrated the importance of 'collaborative capital' across large scale state and national institutions, both which reflect the importance of leaning on networks and sharing knowledge and resources.

Our research continues to observe more abstract characteristics through our case study interviews. The characteristic of being 'opportunistic', for example, was one repeatedly discussed by participants in this case study. Alongside this ability to act on opportunities quickly and wisely, the influence of external actors creating such opportunities needs to also be considered. For the case of the Australian Research Council Linkage grant, it was a non-specific opportunity that had significant influence on the final outcome.

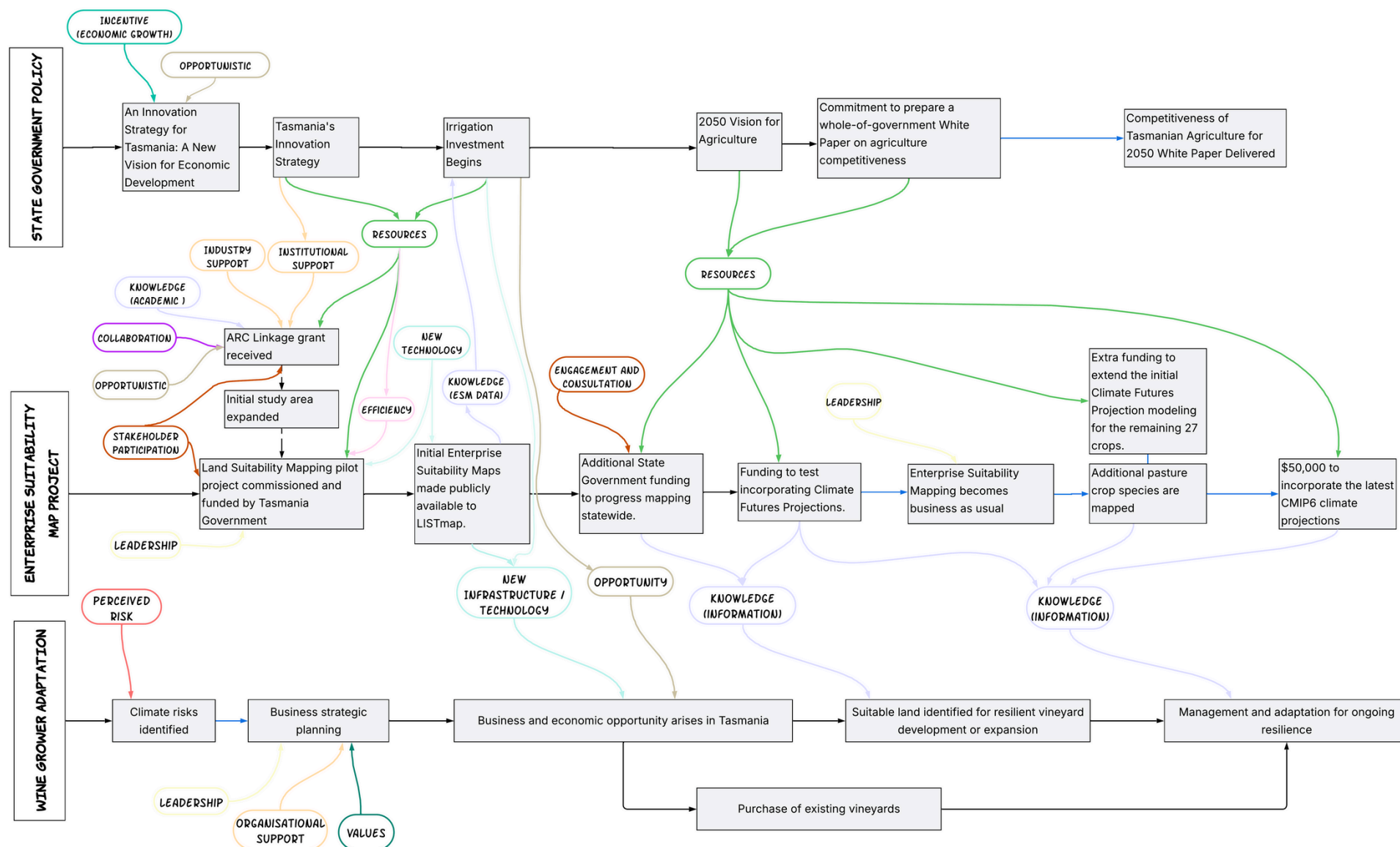


Figure 7: A simplified visualisation of the three processes followed by the Tasmanian Government, the ESM project and wine growers, showing the key enabling factors identified during interviews and their flow from Tasmanian Government down to winegrower

6. Conclusion

This case study provides valuable empirical evidence to demonstrate how enabling adaptation can take shape in complex realities, and how complex enabling environments can be. Among these complexities arise three main lessons from this work.

First, this case study highlights the importance of quality government policies with substantial, ongoing investment and coordinated, well-backed strategies in seeing through substantive change. Continuity in investment to this degree is rare.

Second, this case study suggests that while climate information is extremely important in enabling quality adaptation decisions, it is not a catalyst for change and must be well-timed and well supported by other enabling factors.

Finally, we reiterate the necessity of quality participation and how integral this can be to adaptation outcomes, leading usable information products that fulfil the needs of stakeholders.



Photo: Slovegrove photos

References

1. West J. An Innovation Strategy for Tasmania: A New Vision for Economic Development [Internet]. Hobart, Tasmania; 2010. Available from: <https://nla.gov.au/nla.obj-1382446695/view>
2. Department of Economic Development, Tourism and the Arts. Tasmania's innovation strategy [Internet]. Hobart, Tasmania; 2010. Available from: <https://nla.gov.au/nla.obj-1382446695/view>
3. Department of Natural Resources and Environment Tasmania. Agribusiness Insights: Production Update 2023-24 [Internet]. Hobart, Tasmania; 2024. Available from: <https://nre.tas.gov.au/Documents/Agribusiness%20Insights%202023-24.pdf>
4. Department of Natural Resources and Environment Tasmania. Competitiveness of Tasmanian Agriculture for 2050 [Internet]. 2020. Available from: <https://nre.tas.gov.au/Documents/Competitiveness%20of%20Tasmanian%20Agriculture%202050%20White%20Paper.pdf>
5. Howlett J, Minister for Primary Industries and Water. Research to take irrigation practices to the next level. 2024. Available from: <https://www.premier.tas.gov.au/latest-news/2024/august/research-to-take-irrigation-practices-to-the-next-level>
6. Kidd D, Webb M, Malone B, Minasny B, & McBratney A. Digital soil assessment of agricultural suitability, versatility and capital in Tasmania, Australia. *Geoderma Reg.* 2015;6:7–21.
7. Kidd D, Malone B, McBratney A, Minasny B, & Webb M. Operational sampling challenges to digital soil mapping in Tasmania, Australia. *Geoderma Reg.* 2015;4:1–10.
8. Kidd D, Searle R, Grundy M, McBratney A, Robinson N, & O'Brien L, et al. Operationalising digital soil mapping—Lessons from Australia. *Geoderma Reg.* 2020;23:e00335.
9. Department of Natural Resources and Environment Tasmania. Enterprise Suitability Maps [Internet]. 2024. Available from: <https://nre.tas.gov.au/agriculture/investing-in-irrigation/enterprise-suitability-toolkit/enterprise-suitability-maps>
10. Kidd D, Webb M, Malone B, Minasny B, & McBratney A. Eighty-metre resolution 3D soil-attribute maps for Tasmania, Australia. *Soil Res.* 2015;53(8):932–55.
11. Kidd D, Malone B, McBratney A, Minasny B, & Webb M. Digital mapping of a soil drainage index for irrigated enterprise suitability in Tasmania, Australia. *Soil Res.* 2014;52(2):107–19.
12. Webb M, Kidd D, & Minasny B. Near real-time mapping of air temperature at high spatiotemporal resolutions in Tasmania, Australia. *Theor Appl Climatol.* 2020;141(3):1181–201.

References

13. Webb M, Pirie A, Kidd D, & Minasny B. Spatial analysis of frost risk to determine viticulture suitability in Tasmania, Australia. *Aust J Grape Wine Res.* 2018;24(2):219–33.
14. Webb M, Hall A, Kidd D, & Minansy B. Local-scale spatial modelling for interpolating climatic temperature variables to predict agricultural plant suitability. *Theor Appl Climatol.* 2016;124:1145–65.
15. Department of Primary Industries, Parks, Water and Environment. Enterprise Suitability Maps Metadata (Based on Climate Futures For Tasmania) [Internet]. 2021. Available from: <https://www.thelist.tas.gov.au/app/content/data/geo-meta-data-record?detailRecordUID=e8c81063-2b51-4684-b407-d56d64788a93>
16. Harris R, Remenyi T, Rollins D, Love P, Earl N, & Bindoff N. Australia's wine future-climate information for adaptation to change. 2020.
17. Hayes P. Adaptation of viticulture to climate change in Australia. *El Sect Vitivinícola Frente Al Desafío Cambio Climático.* 2018;117.
18. CSIRO. Alternate winegrape varieties [Internet]. 2024. Available from: <https://www.csiro.au/en/research/plants/crops/horticulture/alternate-grape-varieties>
19. Smith M. Apogee vineyard: Combining old world methods with new world thinking. *Wine Vitic J.* 2013;28(5):74–6.
20. Department of Natural Resources and Environment Tasmania. Tasmania Agri-Food Scorecard Dashboard [Internet]. 2021. Available from: <https://app.powerbi.com/view?r=eyJrIjoiaZDA3YjQxYTctMTEwZC00ZDUwLWl0ZjYtMmQxMWE0ZDYwN2ViliwidCI6IjEwNjNjZmY0LTU5NjYtNGY0YS05MTQ2LWE2ZmRkY2VjMTMxNCJ9>
21. Solecki W, Orlove B, & Bolgert A. The Role of Catalysts in the Climate Adaptation Process. *Wiley Interdiscip Rev Clim Change.* 2025;e931.

Ethics

This project has human research ethics approval from The University of Melbourne (Project ID 29005).

Glossary

ESM: Enterprise Suitability Maps

DNRE: Tasmanian Department of Natural Resources and Environment (DNRE)

NESP: National Environmental Science Program (NESP)

DPIPWE: Department of Primary Industries, Park, Water and Environment

DSM: Digital Soil Mapping

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